Assignment – 11

Power, pivot, solver

1. what is power pivot, and how does it extend the capabilities of excel?

Power pivot is an advanced data modeling and analysis tool in excel that allows users to work with large datasets, create relationships b/w multiple tables, and perform powerful calculations using DAX.

* how power pivot extends the capabilities of excel

1. handles large Data volumes.

* works with millions of rows
* uses in memory data compression for fast performance

2. Create relationships b/w Tables:-

* Connect multiple tables like a database
* No need for complex VLOOKUP or XLookUP.

3. Advanced calculations using DAX:-

* Create calculated column and measures
* perform time intelligence.

4. Powerful Data Analysis:-

* enables advanced pivot tables and pivot charts
* Analyze data from multiple sources together.

5. multiple Data sources:-

* Import data from databases, csv files, text files and more
* Combine data easily for analysis

6. Improved performance:-

* Faster calculations compared to normal excel formulas

* efficient memory usage.

ex:- you can connect:-

* sales-table
* customer table
* product-table and analyze total sales by customer & product without lookup formulas.

2, how do you import data into power pivot from multiple sources?

steps to import data into power pivot-

1. enable power pivot

* go to file → options → Add-ins
* At manage : com Add-ins, click go
* click microsoft power pivot for excel → ok.

2. open power pivot window

* go to the power pivot tab
* click manage

3. Import data from the first source

* In power ~~poin~~ pivot window, click home → get external data
* Choose a source, such as:-
  * from a database
  * from Text /csv
  * from excel
* follow the import wizard and load the table.

4, Import data from Additional sources.

* Repeat get external data
* select another data source
* Import the required tables.

5. Create relationships b/w Tables
* Switch to Diagram view in power pivot
* Drag and connect common fields

6. use Data in excel.
* click pivot table → New pivot table
* Analyze combined data from all sources.

ex:-

You can import
1. Sales data from excel
2. customer data from a csv file
3. product data from a sql database

3. Explain the process of creating relationships b/w tables in power pivot

steps to create relationships in power pivot
1. Load Tables into power pivot
* Import all required tables into power pivot
* each table should have a common column (e.g: customer ID, Product ID)
2. open power pivot window
* go to the power pivot tab
* click manage
3. Switch to diagram view.
* In the power pivot window, click diagram view
* You'll see all imported tables as boxes
4. Identify common columns.
* choose a unique key column in one table (primary)
* choose a matching column in another table (foreign)
ex: customers table → customer ID (unique)
Sales table → customer ID (repeated)

5. create the relationshipt

* Drag the key column from one table to the matching column in the other table

* Power pivot automatically creates the relationship

6. verify the relationshipt

* click Design → manage relationships

* check:

  * correct tables & columes

  * relationships type

4. how would you create calculated columns and measures in power pivot.

Creating a calculated column in power pivot

steps:-

1. open power point
   * go to power pivot → manage

2. select the required table

3. scroll to the empty column on the right

4. click the formula bar

5. Enter a DAX formula

6. Press enter

*example:-

TotalAmount = Sales [quantity]+ Sales [price]

* A new column Total Amount is created.

Creating a measure in power pivot.

steps:-

1. open power pivot → manage

2. go to the home tab

3. click new measure

4. Choose the table
5. Enter the measure name and DAX formula
6. Click ok.

ex:- Total sales = SUM(sales [Total-Amount])

* This measure can be used in pivot Tables.

5) what is the purpose of the solver add-in in excel?

The Solver add-in in excel is used to find the best possible solution for a problem by changing one & more variables, while satisfying given constraints what solver does.

Solver can:-

* maximize a value
* minimize a value
* Set a value to a specific number
* work within constraints

* key components of Solver

1. objective cell - The result you want to optimize
2. variable cells - cells solver is allowed to change
3. Constraints - conditions that must be met (≤, ≥, =)

ex:-

Problem:- A company wants to maximize profit by deciding how many units of product A & B to produce with limits on labor and raw materials.

Solver will:-

* change production quantities
* follow resource limits
* give the maximum profit solution

where it is used :-

* Business optimization
* engineering & operations research
* Finance & investment planning
* scheduling & resource allocation.

6) how do you set up constraints in solver to optimize a solution?

steps to set up constraints in solver.

1. enable Solver.

* go to file → options → Add-ins
* At manage :- Excel Add-ins, click go
* check solver Add-in → ok

2. prepare your worksheet.

* objective cell :- cells that calculates the result

* variable cell :- cells solver can change
* constraint cell :- cells with limits & rules.

3. open solver

* go to data tab → solver

4. set the objective.

* In set objective, select the objective cell

* choose :-
  * max
  * min
  * value of

5. Define variable cells

* In By changing variable cells, select the decision variable cells.

6) Add Constraints :-

* click Add
* In the Add constraint dialog :
    * Cell reference :- cell to constrain
    * operator : $\leq$, $\geq$, =, int, bin
    * constraint :- Limit or condition
* click Add to enter more constraints, or, ok to finish

7) Choose Solving method :-

* Simplex LP → Linear problem
* GRG nonlinear → Nonlinear problems
* Evolutionary → Complex models

8) Solve the model :-

* click Solve
* choose keep Solver solution
* click ok.

* Example constraints
* production $\leq$ 100 units
* Cost $\leq$ ₹ 50,000
* units $\geq$ 0
* Integer values only
* Conclusion :

* Constraints in Solver define the rules and limits for the optimization problem. By correctly setting them, Solver finds the best possible solution that satisfies all conditions.